The Essence of the Cellular Automaton

An emergent spiral on a pulsating wavefront

1 State per cell

Each cell of the L^3 lattice ($L = 321$) stores only:

$$\text{cell} \;=\; (\, r^2,\;\; r,\;\; \texttt{active},\;\; \texttt{spin}\,),$$

where r^2 is the squared Euclidean distance to the center ($\texttt{INF}$ = not yet visited), r is the integer radius, and two bits mark membership: `active` (on the pulsating shell) and `spin` (on the spiral arm).

2 Pulsating wavefront — the "sum of odds"

The distance to the center is never computed by a square root or by multiplication: it propagates through BFS. A unit step along axis i changes r^2 by exactly the corresponding odd number, because

$$(n+1)^2 - n^2 \;=\; 2n + 1. \tag{1}$$

Hence, for a cell x and its neighbor $x + e_i$, where c_i denotes the coordinate centered on axis i:

$$r^2(x + e_i) \;\leftarrow\; \min\!\left(r^2(x + e_i),\;\; r^2(x) + 2\,|c_i| + 1\right). \tag{2}$$

On top of this field acts the pulse: a *triangular* wave in r^2 space,

$$p(t) \in \left[\,0,\;\; 0.92\,R_{\max}^2\,\right], \tag{3}$$

which expands and contracts over time. A cell belongs to the living shell exactly when

$$\texttt{active}(x,t) \;\iff\; \left|\,r^2(x) - p(t)\,\right| \;\le\; \tau, \tag{4}$$

with a small tolerance τ (≈ 1).

3 The helical walker (3D Bresenham around an arbitrary axis)

3.1 Target geometry

A cylinder of radius R_{cyl} whose axis is the line $\{\,P + sA\,\}$, with

$$|A| = \tfrac{L}{2}, \qquad |P| = R_{\text{cyl}}, \qquad P \perp A. \tag{5}$$

Because the center lies on that cylinder surface, the helix starts at $r = 0$ and ends at $r = L/2$ (polar angle $\theta : 0 \to \pi$): one full half-turn while climbing from the core to the outer shell.

3.2 Radius to the axis without multiplication

For $u = v - P$ (offset from the cylinder axis line), the squared distance to the axis is

$$d^2 \;=\; \frac{|u \times A|^2}{|A|^2}, \tag{6}$$

so the walker compares $|u \times A|^2$ against the fixed target $\mathrm{TGT} = R_{\mathrm{cyl}}^2 |A|^2$. Both the cross product $c = u \times A$ and its squared norm $E = |c|^2$ are maintained *incrementally*: a lattice move e_m changes c by the constant vector

$$D_m \;=\; e_m \times A, \tag{7}$$

and changes E by

$$G[m] \;=\; 2\,(c \cdot D_m) + |D_m|^2, \tag{8}$$

whose own update after move k uses the precomputed table

$$K[k][m] \;=\; 2\,D_k \cdot D_m. \tag{9}$$

This is the exact generalization of the classic "$2dx + 1$" increment from scalar coordinates to cross-product space.

3.3 Orbital step

Among the six lattice moves whose tangential projection is positive ($t = -c$ enforces a single sense of rotation), pick the one minimizing the radial error

$$\left|\, E + G[m] - \mathrm{TGT} \,\right|, \tag{10}$$

restricted to the tolerance band $|\cdot| \le \mathrm{TOL}$ (about one cell); ties favor the largest angular progress. If no move stays inside the band, fall back to minimum radial error.

3.4 Climb step

A multi-dimensional DDA along the axis: each accumulator obeys $dda_i \mathrel{+}= |a_i|$; the largest one fires the climb move and then subtracts $|a_x| + |a_y| + |a_z|$. Lattice moves are therefore distributed exactly in proportion to the axis components.

3.5 Interleaving

A Bresenham accumulator interleaves orbital and climb steps with period

$$\mathit{orbit\_total} + \mathit{climb\_total}, \tag{11}$$

tracing the helix one cell per tick.

4 The constraint that defines the automaton's character

At runtime only additions, subtractions, shifts, and comparisons exist: *no multiplication, no division, no floating point, no trigonometry.* Everything that depends on the chosen axis — D_m, $K[k][m]$, P, TGT, TOL, *orbit_total* — is derived once at initialization time, where multiplication is allowed. All geometric intelligence lives in precomputed constants plus local comparison rules.

5 Emergence

`spin` draws the arm; `active` paints the living shell. The spiral itself is never programmed: it emerges from the local rule "minimize radial error while keeping the rotation sense," executed on top of the pulsating bubble.

6 This edition: the ant-view broadcast

An additional layer follows the same philosophy. Each cell holds one monotonically growing value (0 = never reached). When the walker computes a new spiral point, that cell receives the current tick as its value. Then, once per tick, every cell inspects *only* its six face neighbors and copies the greatest neighbor value if it exceeds its own:

$$v(x) \;\leftarrow\; \max\big(v(x),\; v(x \pm e_i)\big). \tag{12}$$

No cell ever senses beyond its neighborhood, yet the values diffuse as waves that in finite time reach every cell of the L^3 grid — global coverage achieved purely by local moves, in lockstep with the CA tick.